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Date:

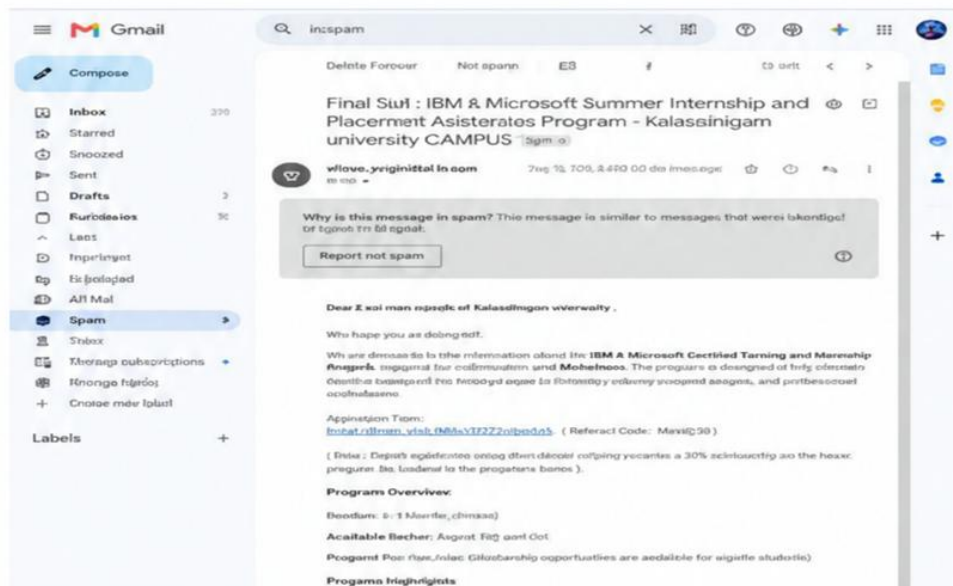
Analyze email headers and detect email spoofing using MHA (Mail Header Analysis)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

Message:	-CAQ6/05isnckBBW/_uakvapeGg/J7P0Bokkwxet.5e1fMaPfoOajIT h@vmat@gmail.com"
Created:	Sun, Aug 16, 2026 at 3:50 PM (Delivered dlier 0 onclesb)
From:	katma.Cgt_dfginal.cdoi
To:	ketraazetf.ongd@cod.com
Subject:	Final Sits : IBM & Microsoft Summer teteraship and Racement Assisterates Preagers - Kalassinigain university CAMPUS
SPF:	PASS with 2000.90.200.41: Learn more
DKIM:	PASS with domain rft.edu: Learn more
DMARC:	PASS: Learn more

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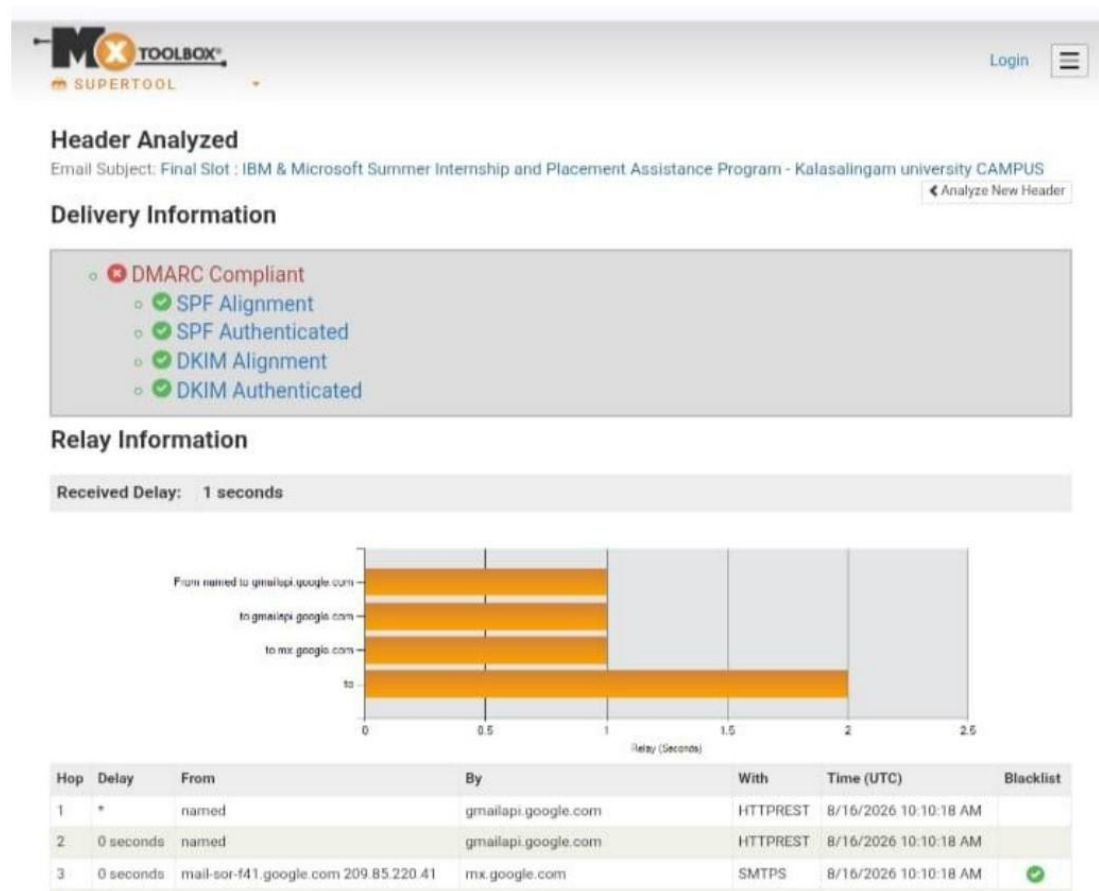
Delivered-To: skrpoeab2000qpa31.com
Received: by skrpoeab2000qpa31.com with SMTP id j3hacp15772kay; Sun, 16 Aug 2026 13:50:21 -0700

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.



5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

SPF and DKIM Information

dmARC:iift.edu Show Solve Email Delivery Problems

```
v=DMARC1; p=none; rua=mailto:email_issues@iift.edu
```

spf:iift.edu:209.85.220.41 Hide Solve Email Delivery Problems

```
v=spf1 ip4:203.190.248.180 include:_spf.google.com include:zcsend.in ~all
```

Prefix	Type	Value	PrefixDesc	Description
Prefix	Typev	Valuespf1	PrefixDesc	DescriptionThe SPF record version
Prefix*	Typeip4	Value203.190.248.180	PrefixDescPass	DescriptionMatch if IP is in the given range.
Prefix*	Typeinclude	Value_spf.google.com	PrefixDescPass	DescriptionThe specified domain is searched for an 'allow'.
Prefix*	Typeinclude	Valuezcsend.in	PrefixDescPass	DescriptionThe specified domain is searched for an 'allow'.
Prefix	Typeall	Value~all	PrefixDescFail	DescriptionThe email matches none of the records.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.

- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.



Whois IP 203.190.248.180

Updated 6 seconds ago

% [whois.apnic.net]
 % Whois data copyright terms http://www.apnic.net/db/dbcopyright.ht
 % Information related to '203.190.248.0 - 203.190.248.255'
 % Abuse contacts for '203.190.248.0 - 203.190.248.255' are 'binalpanda@i

```

inetnum:        203.190.248.0 - 203.190.248.255
netname:        IIFTNET
descr:          Indian Institute of Foreign Trade
country:        IN
admin-c:        PP177-AP
tech-c:         BP201-AP
mnt-routes:     MAINT-IN-BIMAL
mnt-routes:     MAINT-IN-IRINN
remarks:        IIFT Multihoming Network
mnt-by:         MAINT-IN-IRINN
mnt-irt:        IRT-IIFT-AC-IN
status:         ASSIGNED PORTABLE
last-modified:  2025-08-11T22:59:06Z
source:         APNIC

irt:            IRT-IIFT-AC-IN
address:        B-21, Qutab Institutional Area
address:        New Delhi 110016
address:        INDIA
e-mail:         bala@iift.edu
abuse-mailbox:  binalpanda@iift.ac.in
abuse-mailbox:  bala@iift.edu
admin-c:        PP177-AP
tech-c:         BP201-AP
auth:           # Filtered
mnt-by:         MAINT-IN-BIMAL
mnt-by:         MAINT-IN-IRINN
last-modified:  2026-07-27T09:44:58Z
source:         APNIC

person:         Bimal Kuamr Panda
nic-hdl:        BP201-AP
e-mail:         bala@iift.edu
address:        B-21, Qutab Institutional Area
address:        New Delhi 110016
address:        INDIA
phone:          +91-11-26857356
fax-no:         +91-11-26966165
country:        IN
mnt-by:         MAINT-IN-BIMAL
mnt-by:         MAINT-IN-IRINN
last-modified:  2025-09-27T09:36:21Z
source:         APNIC

person:         Pawan Kumar Puri
    
```

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Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Here's a quick example with a sample

header: vbnet Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1]) by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- Check IP Addresses: Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- Verify SPF/DKIM: Check if mail.example.com is authorized to send emails for this domain.
- Compare Time Stamps: Ensure the times make sense chronologically without unusual delays.